FeederFlex Phase 3 — Data Source Audit (Stage 1)

Critical Data Principle

No public dataset represents "a real low-income Indian feeder containing 50-200 households." That does not exist. Every dataset and derived value in this document is labeled:

- **A. REAL** -- directly observed/measured
- **B. DERIVED** -- calculated from real data via a documented transformation
- **C. SYNTHETIC** -- generated by a FeederFlex model, not measured anywhere
- **D. ASSUMED** -- an explicitly chosen modeling parameter with no data source

This is Stage 1 only: dataset audit, verification, selection, evidence, gaps, and an acquisition plan. No canonical schemas, no pipeline code, no synthetic feeder construction -- that is Stage 2 / Phase 4, explicitly not started here (see the closing section).

Dataset Source Audit

Important correction from prior phases: "iAWE" and "Batra et al. dataset" were listed as two separate sources in Phase 1/2. They are **the same source** -- Batra, Gulati, Singh & Srivastava, "It's Different: Insights into Home Energy Consumption in India," ACM BuildSys 2013, is the paper that released the iAWE dataset. This is fixed in the table below; there is no separate "Batra ownership dataset."

Each row below answers the brief's 20 verification questions in condensed form (exists / accessible / what it measures / resolution / Indian / residential / appliance-level / weather / renewable / income / license / transformations needed).

Dataset	Source	Exists/Accessible?	Geography	Years	N	Resolution	Appliance-level?	Weather?	Renewable?	Income info?	License/access	Verified
iAWE	Batra et al., ACM BuildSys 2013, IIIT-Delhi (iawe.github.io)	Yes, exists and publicly documented; download portal referenced in the paper	Delhi, India	2013	**1 household** (not multiple -- corrected from Phase 1/2's implication)	1 Hz electrical, appliance sub-metered (63 appliances labeled for 1 day)	Yes, appliance sub-metered via jPlug/CT sensors	No	No	No	Released "for public use" per the paper; academic citation expected	Confirmed via paper text + GitHub mirror + NILMTK ecosystem references
NEEM Dashboard	Bureau of Energy Efficiency (BEE) + CLASP, neemdashboard.in	Yes, live public government dashboard	13 Indian cities	2018-2020 (153-day window used in one published study)	described in the literature as approximately 200 households metered across 13 cities, plus a 5,000-household ownership/usage survey -- exact accessible count not independently confirmed in this pass; external review flagged a possible 205 in a cited study's sample-allocation appendix, unverified here either way	Hourly	**Ownership/socioeconomic metadata is access-restricted on the public dashboard** per an independent study that used it (confirmed limitation, not assumed)	No (not confirmed as a bundled variable)	No	Restricted (income range not publicly exposed per the same study)	Government source, citation required per site's own terms	Confirmed via BEE/CLASP press material + neemdashboard.in + an independent peer-reviewed study that used and documented its access limits

Dataset	Source	Exists/Accessible?	Geography	Years	N	Resolution	Appliance-level?	Weather?	Renewable?	Income info?	License/access	Verified
Kaggle -- "Electricity smart meter data from Indian two cities" (Mathura, Bareilly)	Community-uploaded, Kaggle	Yes, page exists and is downloadable	Mathura, Bareilly, India	Not independently re-verified this pass	Not independently re-verified this pass	High-frequency smart meter (as titled)	Not confirmed	No	No	No	Kaggle standard terms; uploader-set license -- **[VERIFY IN download: exact license, coverage, missingness]**	Page confirmed to exist (Phase 1 search); contents not re-inspected in this pass
Kaggle -- "Indian Household Electricity Consumption Dataset "	Community-uploaded, Kaggle	Yes, page exists	India, city + DISCOM fields present per listing	Not verified	Not verified	Monthly /appliance-hours (as titled, not sub-15-min)	Appliance *hours of use*, not power draw	No	No	No	Kaggle standard terms	Page confirmed to exist; **not appliance power data, only usage-hours -- do not conflate with Dataset B (appliance profiles)**
Mumbai slum/SRH energy-poverty survey	Peer-reviewed paper, arXiv 2504.17056, "Evaluating energy inefficiency in energy-poor households in India"	Yes, paper exists and is accessible	Mumbai slums + self-reconstructed housing (SRH)	Survey-based, year not confirmed in this pass	Survey sample (exact N not extracted in this pass)	Survey/cross-sectional, not time-series	**Appliance ownership, coded 0/1, by household**, explicitly for low-income housing	No	No	**Yes -- household income in quartiles, explicitly the low-income segment FeederFlex targets**	Academic paper; underlying microdata access not confirmed, only published summary statistics	Confirmed via paper text (abstract + methodology + summary table directly quoted)

Dataset	Source	Exists/Accessible?	Geography	Years	N	Resolution	Appliance-level?	Weather?	Renewable?	Income info?	License/access	Verified
Open-Meteo	open-meteo.com	Yes, live, confirmed free, no API key (Phase 1 s28)	Any coordinates, incl. India	Historical to 1940s + live forecast	N/A (weather grid)	Hourly (interpolatable to 15-min)	N/A	**Yes -- this IS the weather source**	Radiation only (not generation -- conversion required, Phase 2 Part K2)	N/A	Data licensed CC BY 4.0 with attribution; this is separate from API service terms -- free-tier access limits and commercial-vs-non-commercial service conditions must be re-checked at acquisition time, not hard-coded into methodology (wording corrected per external review)	Confirmed live in Phase 1 s28
Grid-India (POSOCO) / NLDC-RLDC real-time pages	Grid Controller of India Ltd, Ministry of Power	Yes, live government source	National + 5 regional grids	Live + historical reports	N/A (grid aggregate)	Varies (some real-time, some daily reports)	No	No	**Generation mix by fuel type at regional/national level -- context only, not feeder-level**	No	Government source, disclaimer on reliance for decisions (per NERLDC's own site)	Confirmed live in Phase 1 s28

Dataset	Source	Exists/Accessible?	Geography	Years	N	Resolution	Appliance-level?	Weather?	Renewable?	Income info?	License/access	Verified
CEA distribution planning documents	Central Electricity Authority	Yes, CEA publishes distribution planning guidance	India, national/state	Various reports	N/A	N/A	No	No	No	No	Government publication	**[VERIFY IN PHASE 3 continuation]** -- the specific claim that CEA documents describe 11kV feeders feeding LT consumers through DTs is standard-practice knowledge, referenced by the external Phase 2 reviewer, but the exact CEA document was not independently pulled and cited in this pass
ERA5-Land (ECMWF)	Copernicus/ECMWF	Yes, established reanalysis dataset	Global, incl. India	1950-present	N/A	Hourly	N/A	Yes	Radiation only, same conversion caveat as Open-Meteo	N/A	Free, Copernicus license	Referenced in Phase 1's POWBAL paper citation; a viable Open-Meteo alternative/cross-check

Rejected candidates (named in the brief, checked, not usable): "CarbonTracker.in" -- referenced only as a source scraped by the POWBAL researchers (Phase 1 s3), not independently confirmed as a publicly self-serve dataset in this pass; treat as **unverified**, not rejected outright, pending a direct check. No dataset claiming to be a "real Indian low-income feeder load profile" was found anywhere in this or prior passes -- confirms Phase 1/2's repeated finding that this specific data does not publicly exist.

Dataset Selection

Label set expanded per external review to distinguish verified from not-yet-verified sources -- PRIMARY, PROVISIONAL PRIMARY, SECONDARY, VALIDATION/CROSS-CHECK, CONTEXT, ASSUMED, REJECTED.

Dataset	Role	Why
iAWE	**SECONDARY** -- appliance-level shape reference, with a sharper limit than previously stated: it provides 73 days of household/environmental sensing but **only 1 day of fully labeled appliance-level observations** for 63 appliances (corrected per external review -- do not read "73 days of appliance behavior" into this source). Appliance signatures drawn from it have limited behavioral generalizability and should be described that way in any write-up	

Dataset	Role	Why
NEEM Dashboard	**PROVISIONAL PRIMARY** (downgraded from PRIMARY per external review)	Still the strongest candidate for the load-profile backbone, but three things must be verified before it can be called PRIMARY: (1) raw hourly data is actually downloadable, not just dashboard-visualized; (2) the exact accessible household count -- literature describes "approximately 200," one external review flagged a possible 205 in the cited study's sample-allocation appendix, which this pass has **not independently confirmed** either way; (3) whether the cited 153-day window is one continuous sequence or separate pre-COVID/deep-COVID periods that should not be treated as continuous. Promotes to PRIMARY only after these pass inspection
Kaggle Mathura/Bareilly smart-meter data	**SECONDARY, PROVISIONAL**	Household-count and resolution not independently re-verified this pass
Kaggle Indian Household Electricity Consumption Dataset	**CONTEXT ONLY**	Usage-hours, not power time series

Dataset	Role	Why
Mumbai slum/SRH energy-poverty survey (arXiv 2504.17056)	**PRIMARY for appliance ownership, Dataset C -- scope corrected per external review**	Provides real, income-stratified appliance-ownership evidence, but strictly for **Mumbai slum and self-reconstructed-housing households**, not "Indian low-income households" generally. Used in this project as scenario-specific evidence for the Mumbai deployment scenario, not a national prevalence estimate
Open-Meteo	**PRIMARY** -- weather/solar input	Confirmed live and free for the data itself (CC BY 4.0), but see the licensing note below -- API *service* tier terms are separate from the *data* license and must be re-checked at acquisition time, not hard-coded into methodology
ERA5-Land	**SECONDARY CROSS-CHECK** (renamed from "VALIDATION" per external review)	Same ECMWF data family as Open-Meteo -- a consistency check, not an independent validation source

Dataset	Role	Why
Grid-India / NLDC-RLDC	**CONTEXT ONLY**	National/regional generation-mix context; explicitly **not** a validation source for the local synthetic PV shape (national solar generation reflects installed capacity, transmission, curtailment and plant availability that have no bearing on a local scenario profile) -- qualitative comparison only
CEA distribution/network-planning documents	**CONTEXT, now with a real citation**	"CEA Electricity Distribution Network Planning Criteria" explicitly documents 11 kV feeders serving consumers through DTs, RMU-backed ring configurations, and dedicated feeders for critical loads -- replaces the prior "standard practice" placeholder with an actual citable source (see Grid/Feeder Context section)
CarbonTracker.in	**REJECTED (for now)**	Not independently confirmed as self-serve accessible
Any dataset claiming to be real feeder-level low-income load data	**REJECTED -- does not exist**	Confirmed absent across four research passes now

Appliance Parameter Evidence Table

Honesty note: this pass did not independently pull iAWE's raw appliance metadata file (rated power, purchase date, star rating per appliance, which the paper says the dataset includes). What follows is what's confirmed from the paper's text and general engineering references, not fabricated precision. Where no source gives a number, it is marked ASSUMED and left as a range, not a false point estimate.

Appliance	Parameter	Value/range	Source	Type	Confidence	Notes
Water pump (household)	Rated power	Not extracted from a specific dataset in this pass	--	ASSUMED	Low	Typical Indian domestic pump literature commonly cites a few hundred W to ~1 kW class, but no specific source was pulled and cited here -- **must be sourced properly in Phase 3 continuation, not carried forward as a number**
Geyser/water heater	Rated power	1,000-2,000 W	Phase 1's POWBAL paper citation (Khanna et al., Tata Power trial), which reports this range for Indian electric geysers	REAL (cited from a specific study)	Medium -- from a real cited Indian trial, not this project's own measurement	Same source found geyser switch-offs showed no measurable household-level effect, implying strong on-demand usage resistance to curtailment

Appliance	Parameter	Value/range	Source	Type	Confidence	Notes
Refrigerator	Rated power	100-200 W (full-power operation)	Same POWBAL paper citation	REAL (cited)	Medium	POWBAL found refrigerators had the largest measurable curtailment effect of the three appliance types they tested
Room AC	Rated power	1,500-2,500 W depending on capacity/efficiency	Same POWBAL paper citation	REAL (cited)	Medium	Not FeederFlex's primary target appliance (Phase 1 s14), included for completeness since it's the best-evidenced appliance in the cited literature
Cooler	Rated power	Not sourced in this pass	--	ASSUMED	Low	**Gap -- flagged for Phase 3 continuation, not fabricated**
Appliance ownership -- pump, geyser, cooler, fridge, AC (low-income Mumbai households specifically)	Ownership prevalence	Coded 0/1 per household, quartile-stratified by income; exact percentages not extracted from the summary table in this pass beyond fridge (~90%) and AC (~2.2%)	Mumbai slum/SRH energy-poverty survey, arXiv 2504.17056	REAL (survey, cited)	Medium-high -- directly on-target population, but only two appliance rates were pulled from the visible summary table in this search pass	**The single most valuable finding of this audit** for grounding the "low-income" framing in actual data rather than assumption -- full extraction of the paper's complete ownership table is a Phase 3 continuation task

Appliance	Parameter	Value/range	Source	Type	Confidence	Notes
Interruption/flexibility windows (any appliance)	Max interruption, cooldown, minimum runtime	No dataset found that measures this directly for Indian households	--	ASSUMED	Low	This was already flagged as an assumption-heavy area in Phase 2 (Part AD); this Phase 3 pass did not close that gap -- no literature source for these specific behavioral parameters was located

Bottom line: real, cited evidence exists for rated power of geyser/fridge/AC (via the POWBAL citation already used in Phase 1) and for appliance *ownership* in a genuinely low-income Indian population (the new Mumbai survey find). It does **not** exist for pump/cooler rated power or for any appliance's interruption/cooldown/minimum-runtime behavior -- these remain engineering assumptions (Phase 2 Part AD) that Phase 3 continuation should either source properly or explicitly carry forward as labeled ASSUMED parameters, never presented as measured.

Household Diversity and Low-Income Labeling

Can the available data support heterogeneity, not 200 identical households? Partially. NEEM's 200-household/13-city sample gives real diversity in daily consumption, peak demand, and dwelling type (the earlier-cited lockdown study specifically used RM1/BR1/BR2/BR3/M3BR dwelling-typology labels from NEEM). iAWE gives real within-household appliance-level shape but only one household's shape. The Mumbai energy-poverty survey gives real cross-household variance in consumption and ownership specifically within a low-income population, but as survey/cross-sectional data, not a time series -- it can inform *which* appliances a synthetic household is likely to own and roughly how much it consumes annually, not its 15-minute load curve.

Combining these without pretending they're one dataset: the intended Phase 4 approach (flagged here for Phase 3 to prepare for, not executed here) is to use NEEM-derived load *shapes* as the time-series backbone, iAWE-derived appliance *shapes* to disaggregate those into per-appliance behavior, and the Mumbai survey's ownership

probabilities to decide which synthetic households get which appliances -- three different real sources, each doing a different job, combined through documented assumptions, not merged as if they were one coherent dataset.

Low-income labeling verdict, scope tightened per external review: the synthetic feeder must not be called "a low-income dataset" outright, and the Mumbai slum/SRH survey specifically must not be generalized past its actual population. This audit found one genuinely low-income-specific source (the Mumbai survey) that supports real ownership-probability grounding **for that Mumbai deployment scenario specifically**, not for "Indian low-income households" as a national category. No source provides low-income-specific *time-series load curves*. Corrected framing: **"a synthetic feeder representing a low-income/peri-urban deployment scenario (Mumbai slum/SRH-informed), using observed Indian household consumption shapes and, where available, Mumbai-specific income-stratified appliance-ownership evidence."** Household income itself remains unmodeled as a continuous variable -- only ownership *probability*, for that one city's low-income population, is informed by income-stratified data.

Temporal resolution mismatch, flagged per external review as the most important technical issue in this document: NEEM provides **hourly** data; Phase 2's forecasting design uses 15-minute lags including t-96 (same time yesterday). Interpolating hourly to 15-minute does not create real 15-minute measurements. The corrected approach:

```
NEEM hourly demand envelope          [REAL]
        |
        v
15-minute intra-hour shape           [DERIVED/SYNTHETIC -- from a
(from a documented high-resolution     documented high-resolution load-
load-shape source, not invented)       shape reference, e.g. iAWE's
        |                              appliance-level sub-hourly pattern]
        v
15-minute synthetic household profile
```

Any 15-minute figure downstream of NEEM must be labeled DERIVED/SYNTHETIC, never presented as an observed 15-minute measurement, and its sensitivity to the disaggregation method should be checked before Phase 4 relies on it heavily. **Separately:** the cited 153-day NEEM window spans pre-COVID (2018-2019) and deep-COVID (2020) periods per the source study -- this must not be treated as one continuous sequence for

lag-feature purposes (t-96 assumes continuity) until timestamp continuity within each period is directly verified; the periods should be treated as separate sequences unless proven otherwise.

Appliance disaggregation must conserve energy -- added per external review, a genuine methodological gap in the prior version of this document. Combining NEEM (household aggregate), iAWE (appliance shapes), and the Mumbai survey (ownership) is not just a matter of using each for a different purpose -- naively adding appliance-level curves on top of an aggregate load would double-count energy. The mandatory Phase 4 constraint:

```
NEEM household aggregate load
        |
        v
ownership constraints (Mumbai survey -- which appliances this synthetic
household has)
        |
        v
iAWE appliance shape library (how each owned appliance's load is shaped over
time)
        |
        v
appliance allocation / disaggregation
        |
        v
residual "other loads" (whatever the aggregate doesn't attribute to a modeled
appliance)
        |
        v
sum(allocated appliance loads + residual) MUST equal the NEEM household
aggregate load
```

This equality is a hard constraint on Phase 4's synthetic feeder builder, not a suggestion -- flagged here so it can't be silently skipped when implementation begins.

Weather and Renewable Supply Data

Weather (Dataset D): Open-Meteo is confirmed real, live, free, and India-covering (Phase 1 s28, re-confirmed here). Useful variables for FeederFlex, and only these, per the brief's instruction not to include variables merely because they're available: temperature (cooling-load proxy, feeds Part K demand forecast), shortwave/solar radiation (feeds Part

K2′s renewable conversion model), cloud cover (secondary radiation-quality signal). Humidity and precipitation are available but not clearly tied to any FeederFlex model component -- **not included** unless a specific use is identified later, per the brief′s "don′t include merely because available" instruction. Used for **both** load forecasting (temperature) and renewable generation modeling (radiation), per Part 11′s question.

Renewable supply (Dataset E): confirmed, again, that no real local/feeder-level renewable generation dataset exists publicly. Solar radiation from Open-Meteo is not generation. The only defensible path is a **DERIVED/SYNTHETIC renewable generation profile**, computed from real Open-Meteo radiation via a documented PV conversion formula. **Variable named precisely per external review:** the input is specifically Open-Meteo′s shortwave solar radiation variable (Global Horizontal Irradiance, GHI, in W/m2), not generic "solar radiation":

```
PV_output_kW = PV_capacity_kW * performance_ratio * (GHI_Wm2 / 1000)
  clipped at PV_capacity_kW
```

with `PV_capacity_kW` an explicit ASSUMED scenario parameter and `performance_ratio` a cited generic PV-derating constant (~0.75-0.85), not an India-specific measured value. **Wording corrected per external review:** Grid-India′s regional generation-by-fuel-type data may be used for *qualitative contextual comparison* with regional/national solar-generation patterns only -- it is explicitly **not** validation of the local synthetic PV profile, since national generation reflects installed capacity, transmission, curtailment, and plant availability that have no bearing on a local scenario′s shape or magnitude. This must be labeled DERIVED/SYNTHETIC in every downstream use, never as "renewable generation data."

Added per external review -- this is what actually makes the project about intermittency, not just a deterministic solar curve: the renewable profile needs both an *actual* profile and, wherever forecasting is evaluated, a separate *forecast* profile with forecast error, mirroring the demand side′s forecast-vs-actual structure (Phase 2 Part K). A deterministic PV curve alone does not model intermittency -- only the actual/forecast/error structure does. Phase 4 must also fix, as explicit scenario parameters rather than leaving implicit: whether generation is behind-the-meter or upstream of the households (this project treats it as **upstream**, per Phase 2 Part K2′s own scope clarification), whether export beyond local demand is modeled at all, and whether renewable availability is treated as deterministic or given explicit uncertainty.

Grid / Feeder Context Data

No public feeder-level (11kV or otherwise) residential load data was found, confirming Phase 1 and Phase 2's repeated conclusion across three independent research passes now. This is stated explicitly rather than left implicit, per the brief's requirement.

What IS available, for context and model parameters only, never as a stand-in for measured feeder data:

- RDSS/NFMS confirms feeder-level and DT-level smart metering is being actively deployed by India's DISCOMs (Phase 1 s15, s28) -- evidence that the 11kV-feeder-through-DTs structure (Phase 2 Part B item 8) is how the real system is organized, even though no data from it is public.
- Grid-India/NLDC-RLDC gives national and regional demand, generation mix, and frequency -- useful for narrative/contextual framing ("here's what net load looks like nationally") and for validating the *shape* of a synthetic renewable profile, never as feeder input.
- **Resolved this pass, per external review's push to get a real citation instead of "standard practice":** the Central Electricity Authority's "Electricity Distribution Network Planning Criteria" document explicitly describes 11 kV feeders serving consumers through distribution transformers, including ring-back-up configurations via RMUs and dedicated feeders for critical loads (hospitals, water works, etc.). CEA's Distribution Perspective Plan separately reports India had 230,979 11 kV feeders as of March 2022, projected to reach 323,899 by 2030. Phase 2 Part B's "VERIFY IN PHASE 3" placeholder can now be replaced with this citation. **Not yet done in this pass:** pulling the exact document URL/version into a formal citation entry for the Phase 3 source registry -- a small remaining task, not a research gap.

National statistics are not feeder measurements, and nothing in this document treats them as such.

Data Compatibility Matrix (NEW, added per external review)

Why this exists: before combining sources, the specific mismatch between each pair and how it will be reconciled needs to be stated -- not just "these serve different purposes," which was this document's prior, insufficiently rigorous framing.

Source	Geography	Population	Resolution	Purpose
NEEM	13 Indian cities	Residential, mixed income (skews upper/middle per the cited study's own note)	Hourly	Aggregate load backbone
iAWE	Delhi	1 household	1 Hz (1 day fully appliance-labeled)	Appliance-level shape reference
Mumbai survey	Mumbai	Low-income (slum/SRH)	Cross-sectional survey, not time-series	Appliance ownership probability
Open-Meteo	Chosen scenario coordinates	N/A (weather grid)	Hourly	Weather input, PV conversion input

Dataset A	Dataset B	Shared dimension	Mismatch	Reconciliation method
NEEM	iAWE	Household load composition	NEEM is aggregate-only; iAWE is appliance-resolved but from one non-target-income household	iAWE supplies *shape* (how a pump/geyser's load looks over its cycle), scaled and allocated within NEEM's aggregate via the conservation-of-energy constraint above -- never used to add new energy on top of NEEM

Dataset A	Dataset B	Shared dimension	Mismatch	Reconciliation method
NEEM	Mumbai survey	Household population	NEEM skews upper/middle income; Mumbai survey is specifically low-income	Mumbai survey is used only to set appliance-ownership *probability* for synthetic households, not to adjust NEEM's load magnitude -- the income mismatch is not resolved by reweighting NEEM, it's handled by treating ownership and load-shape as separate model layers
NEEM	Open-Meteo	Temporal resolution and geography	NEEM is hourly and tied to its 13 surveyed cities; Open-Meteo is available at any resolution/location	Open-Meteo is queried for the chosen scenario's coordinates and resampled to match NEEM's hourly cadence before either is further disaggregated to 15-minute
iAWE	Mumbai survey	Neither shares a household or a city	No direct overlap	These are never merged directly -- iAWE contributes shape, Mumbai survey contributes ownership probability, applied to different synthetic households independently

Legitimacy of combining these sources rests on this table, not on the sources being similar -- they aren't, and this matrix is what makes the combination defensible: each source contributes exactly one layer (aggregate magnitude, appliance shape, or ownership probability), never two, and no pair is asked to agree on something they weren't designed to measure together.

Data Gaps Summary

Gap	Severity	Transparent modeling response (not a fabrication)
No real feeder-level (11kV) residential load data, anywhere, for any income segment	High, but expected and already scoped around in Phase 2	Synthetic feeder built from real household-level data (Phase 4), always labeled SYNTHETIC
No real renewable generation data at feeder/local scale	High	DERIVED/SYNTHETIC PV conversion from real Open-Meteo radiation (Phase 2 Part K2), labeled as such
iAWE is one household, not a diverse sample	Medium	Use NEEM's 200-household/13-city sample as the primary diversity backbone; use iAWE only for within-household appliance *shape*
NEEM's appliance-ownership/socioeconomic layer is access-restricted	Medium	Substitute the Mumbai low-income survey (arXiv 2504.17056) for ownership *probability*, not NEEM
Pump and cooler rated power not sourced	Medium	Left as ASSUMED ranges pending Phase 3 continuation; not fabricated as point values

Gap	Severity	Transparent modeling response (not a fabrication)
No source for interruption/cooldown/minimum-runtime behavior for any appliance	Medium-high -- this feeds directly into Phase 2's DR controller constraints	Remains an explicit ASSUMED/engineering parameter (Phase 2 Part AD). **Wording corrected per external review:** since FeederFlex runs no real field trial on real households, these will be specified as **documented, conservative engineering assumptions and evaluated through sensitivity analysis** (e.g. scenario sweeps across conservative/moderate/permissive interruption and cooldown settings, per Phase 2 Part W), not described as "empirically tuned" -- that word implies a measured calibration this project cannot claim. Per external review, these should also be **appliance-specific**, not one generic constraint set: POWBAL's own findings (Phase 1 s3) already show geysers tolerate far less curtailment than refrigerators or ACs, so a single max-interruption/cooldown pair for "all flexible appliances" would contradict evidence already in hand. Each appliance category needs its own flexibility class.
Two Kaggle datasets not independently re-verified for resolution/quality/license in this pass	Medium	Flagged for direct download-and-inspect in Phase 3 continuation before being relied on

Gap	Severity	Transparent modeling response (not a fabrication)
Specific CEA 11kV/DT citation not pulled	Low-medium -- affects citation rigor, not the architecture's validity	VERIFY IN PHASE 3 continuation; the underlying engineering claim is standard practice regardless
CarbonTracker.in accessibility unconfirmed	Low	Not used until verified; not currently load-bearing for anything in this spec

Recommended Acquisition Plan

Reordered per external review to prioritize the intended load backbone (NEEM) first, since everything else depends on knowing whether it's actually usable before investing further effort elsewhere:

1. **NEEM raw-data access verification** -- confirm whether the 200/205-household hourly series is downloadable (not just dashboard-visualized), the exact accessible count, and whether the 153-day window's pre-COVID/deep-COVID periods are internally continuous. This gates the PROVISIONAL PRIMARY -> PRIMARY promotion.
2. **Kaggle Mathura/Bareilly verification** -- potential backup/additional household diversity if NEEM access is limited.
3. **Mumbai low-income survey -- pull the complete appliance-ownership table**, not just the fridge/AC rates extracted so far: pump, geyser, cooler, washing machine, and any other reported appliance, ideally with income-quartile stratification preserved. **Mandatory, not optional**, per external review -- the synthetic feeder's ownership model depends on it.
4. **iAWE raw data + metadata** -- confirm the download link is live and pull the actual appliance metadata (rated power, star rating) the paper describes, keeping the 1-day-labeled limitation explicit.
5. **CEA feeder architecture citation** -- now largely resolved this pass (see Grid/Feeder Context); remaining work is formalizing the exact document URL/version into the source registry.

6. **Weather acquisition** -- Open-Meteo primary, ERA5-Land as a secondary cross-check; re-confirm current API service-tier terms at acquisition time rather than relying on this document's snapshot.
7. **Pump and cooler power evidence** -- source from BEE star-labeling data, manufacturer spec sheets for representative Indian models, or academic literature; produce min-typical-max ranges with sources cited for each bound, not a single invented number.
8. **CarbonTracker.in** -- only pursued if steps 1-7 leave a genuine gap it could fill; not currently load-bearing for anything in this spec.

This plan is scoped to closing the gaps identified above, not to re-running the entire Stage 1 audit.

STOP -- Stage 1 Complete, Stage 2 Not Started

Per the brief's own instruction, this document stops here. **Not yet done, and deliberately not started in this pass:** canonical schemas (household/appliance/weather/renewable tables), the data pipeline design (ingestion -> validation -> cleaning -> standardization -> canonical dataset), the directory structure, unit/time standardization tables, data quality/missing-data/outlier policies applied to actual downloaded files, the data lineage table, the Phase 4 input contract, the Phase 3 assumption and risk registers, and the final validation checklist -- all of Parts F onward in the original Phase 3 brief's structure remain Stage 2 work, to begin only once the acquisition plan above has actually been executed (datasets downloaded and inspected, not just identified).

Ready for Stage 2 when you are -- it should start from item 1 of the acquisition plan, not from writing pipeline code.

External Review Response -- v1.0 to v1.1 Changelog

A second external (ChatGPT-generated) review raised 21 numbered points against this Stage 1 audit. As with the Phase 2 review, each was checked before acting on it.

All 5 of the review's stated "mandatory" changes were valid and are applied: NEEM downgraded from PRIMARY to PROVISIONAL PRIMARY pending raw-data verification; the 200-vs-possible-205 household count discrepancy is now stated as unverified rather than asserted as 200; the hourly-to-15-minute gap is now explicit with a labeled DERIVED/SYNTHETIC conversion step; the NEEM/iAWE/Mumbai reconciliation now has a hard energy-conservation constraint; "tuned empirically" is replaced with "documented engineering assumptions plus sensitivity analysis."

Also applied, all independently reasonable: iAWE's 1-day-fully-labeled limit stated precisely; the 153-day non-continuity caution; ERA5-Land relabeled a cross-check, not validation; Grid-India explicitly not used to validate the local PV shape; Open-Meteo's data-license-vs-API-terms distinction; the GHI variable named precisely with a clipped conversion formula; the actual/forecast/error renewable structure (this is what makes the project genuinely about intermittency, not a static solar curve); the Mumbai survey rescoped to "Mumbai low-income," not "Indian low-income"; appliance-specific (not generic) flexibility classes; the new Data Compatibility Matrix section; the reordered, NEEM-first acquisition plan.

Handled with an actual resolution, not just a wording tweak: point #20 asked for a real CEA citation instead of "standard practice" -- a live search in this pass found and cited CEA's "Electricity Distribution Network Planning Criteria" document plus CEA's Distribution Perspective Plan feeder counts, resolving the placeholder with a genuine source rather than removing the claim.

Flagged but not independently confirmed: the specific "205 households" figure the review attributed to the cited study's appendix was not verified against that appendix directly in this pass -- the document now says the count is unverified either way, rather than adopting "205" as if this pass had confirmed it. That distinction matters: agreeing a number needs verification is not the same as verifying it.